

Integrated study overview and working model

A Study design and evidence framework



Discovery:
GSE12251
(n=23)



Replication:
GSE16879
(n=24)



Cohort 3:
GSE23597
(n=45)

Pretreatment UC mucosa • Infliximab • GPL570

63-gene cross-cohort consensus signature

Interpretability

Specificity

Localization

Leave-one-dataset-out (LODO) axis

Held-out scoring

Leakage-free

Effect-size evaluation

LODO pooled Hedges' g
0.86 (95% CI 0.28–1.45)
• $p=0.0039$ • $I^2=39\%$

Hallmark pathway convergence

TNF-alpha/NF-kB

IL-6/JAK/STAT3

Interferon

Hypoxia

EMT

Responders: Oxidative phosphorylation

Specificity benchmarks

Inflammation proxy

EMT proxy

Public single-cell localization support

GSE116222 (epithelial)

GSE125527 (rectal immune)

B Working model



Rectal myeloid (M/DC)

TNF-alpha/NF-kB

IL-6/JAK/STAT3

Interferon

Hypoxia

Pretreatment
nonresponse
axis

Association
+ localization



Epithelial remodeling

Cycling/stem-like

Stress-associated

EMT/remodeling

Boundary: association + localization (public data); no causality; no deployable clinical predictor.